DS210: Statistical Modelling Tools

Rational

In today's data-driven world, the ability to harness the power of data through statistical modeling is essential for making informed decisions. This course acknowledges the growing demand for skilled professionals who can analyze data and develop meaningful models across different industries. By integrating a diverse set of statistical modeling tools, it ensures that students are well-prepared to tackle a wide range of real-world challenges where data analysis and modeling are critical.

Learning outcomes

By the end of the course-tenure the students should be able to:

• have a deep understanding of advanced statistical modeling techniques.

• be familiarized with a range of statistical modeling tools commonly used in different industries.

• be equipped with the ability to analyze data, develop models, and make data-driven decisions.

• have critical thinking, problem-solving, and ethical considerations in data analysis and modeling.

• have practical skills through a capstone project that applies learned techniques to real-world scenarios.

• be current with emerging technologies and trends in statistical modeling.

Course content

Unit 1: Introduction to Statistical Modeling and SAS

- What is statistical modeling?
- Introduction to SAS as a statistical modeling tool.
- Types of statistical models.
- Setting up the SAS environment.
- The history and evolution of statistical modeling.
- Understanding the importance of data quality.
- Overview of data sources and data collection methods.
- The role of data governance in statistical modeling.

Unit 2: Data Preparation and Exploration with SAS

- Data cleaning and preprocessing using SAS.
- Exploratory data analysis (EDA) with SAS.

- Data visualization and summary statistics with SAS.
- Handling missing data and outliers using SAS.
- Data transformation techniques in SAS.
- Data scaling and standardization in SAS.
- Working with unstructured data in SAS.
- Practical tips for effective data preparation in SAS.

Unit 3: Descriptive Statistics and Probability with SAS

- Descriptive statistics using SAS.
- Probability distributions with SAS.
- Sampling and random variables with SAS.
- Confidence intervals and hypothesis testing in SAS.
- Interpreting and communicating results from descriptive statistics.
- Advanced probability distributions and their applications.
- Bayesian vs. frequentist approaches to probability.
- Real-world case studies in hypothesis testing.

Unit 4: Regression Analysis in SAS

- Simple and multiple linear regression in SAS.
- Logistic regression with SAS.
- Model evaluation and diagnostics in SAS.
- Polynomial regression and regularization techniques using SAS.
- Generalized linear models (GLMs) in SAS.
- Time-series regression modeling with SAS.
- Nonlinear regression modeling in SAS.
- Model selection and feature engineering in regression analysis.

Unit 5: Time Series Analysis in SAS

- Introduction to time series data.
- Time series decomposition using SAS.
- Forecasting techniques with SAS.
- ARIMA modeling and seasonal decomposition with SAS.
- Exponential smoothing methods using SAS.
- GARCH models for volatility forecasting.
- Machine learning-based time series forecasting.
- Advanced time series analysis case studies.

Unit 6: Statistical Analysis with SPSS

- Introduction to SPSS for statistical analysis.
- Regression analysis in SPSS.

- Correlation and factor analysis in SPSS.
- Real-world applications of SPSS in social science research.
- Advanced statistical techniques in SPSS.
- Survey data analysis with SPSS.
- Predictive modeling with SPSS.
- Automation and scripting in SPSS.

Unit 7: Data Analysis and Modeling with STATA

- Introduction to STATA for data analysis and modeling.
- Data management in STATA.
- Statistical modeling techniques in STATA.
- Applications in economics, sociology, and political science research.
- Survival analysis with STATA.
- Structural equation modeling (SEM) in STATA.
- Multilevel modeling in STATA.
- Real-world case studies and publications using STATA.

Unit 8: Numerical Modeling with MATLAB

- Introduction to MATLAB for numerical and mathematical modeling.
- Using MATLAB for statistical modeling functions.
- Practical applications of MATLAB in modeling.
- MATLAB programming and automation.
- Solving complex mathematical problems with MATLAB.
- Real-world simulations and modeling using MATLAB.
- Data-driven modeling with MATLAB.
- Integration of MATLAB with other software and tools.

Unit 9: Basic Statistical Modeling with Excel

- Overview of Excel for basic statistical analysis and modeling.
- Using Excel for simple statistical modeling tasks.
- Case studies for statistical analysis in Excel.
- Advanced Excel functions for statistical modeling.
- Visualizing data in Excel.
- Data analysis tools and add-ins in Excel.
- Building decision support systems in Excel.
- Spreadsheet automation and macros for modeling.

Unit 10: Advanced Predictive Analytics with IBM SPSS Modeler

- Introduction to IBM SPSS Modeler for predictive analytics.
- Data mining and machine learning using IBM SPSS Modeler.

- Building predictive models and making data-driven decisions.
- Automation and deployment of models with SPSS Modeler.
- Text analytics and natural language processing with SPSS Modeler.
- Advanced model evaluation and optimization.
- Integration with other IBM analytics tools.
- Case studies in predictive analytics and real-time decision-making.

Unit 11: Data Visualization and Analysis with JMP

- Introduction to JMP for data visualization and statistical analysis.
- Interactive data exploration capabilities in JMP.
- Case studies and practical applications of JMP.
- Data visualization principles and best practices.
- Interactive dashboards and reporting in JMP.
- Advanced graphing and visualization techniques in JMP.
- Exploratory data analysis with dynamic graphics.
- JMP scripting and automation for data visualization.

Unit 12: Quality Improvement and Statistical Analysis with Minitab

- Introduction to Minitab for quality improvement and statistical analysis.
- Statistical tools and techniques in Minitab.
- Real-world applications in industries such as manufacturing and engineering.
- Design of Experiments (DOE) with Minitab.
- Process capability analysis using Minitab.
- Quality control and Six Sigma methodologies with Minitab.
- Root cause analysis and problem-solving with Minitab.
- Automation and integration of Minitab in production processes.

Unit 13: Bayesian Statistics

- Bayes' theorem and Bayesian inference.
- Bayesian modeling using MCMC (Markov Chain Monte Carlo).
- Bayesian regression and hierarchical models.
- Practical applications in Bayesian statistics.
- Bayesian networks and graphical modeling.
- Bayesian model selection and model comparison.
- Case studies in Bayesian data analysis.
- Ethical considerations in Bayesian modeling.

Unit 14: Big Data and Distributed Computing

- Handling and analyzing large datasets.

- Introduction to distributed computing (e.g., Apache Spark).
- Parallel processing and distributed modeling.
- Scalability and performance considerations.
- Hadoop and MapReduce.
- Data storage and retrieval in big data environments.
- Real-time data processing and streaming analytics.
- Machine learning on big data platforms.

Teaching and Learning Methods

Lectures:90Hrs Tutorials:60Hrs, Labs:60Hrs, Assessments:30Hrs, NATIONAL HRS: 240

Assessment

- Continuous Assessment 40%
  - 20% for hands-on assessments
  - 20% for non-hands-on assessments
- Final Exams 60%
  - 20% for hands-on assessments
  - 30% for theory assessments

Recommended Readings

1. Cody, R. P., & Smith, J. K. (2006). *Applied statistics and the SAS programming language* (5th ed.). Pearson.
2. Bowerman, B. L., O'Connell, R. T., & Orris, J. B. (2013). *Essentials of business statistics with applications in SAS* (5th ed.). McGraw-Hill.
3. Shumway, R. H., & Stoffer, D. S. (2017). *Time series analysis and its applications with R examples* (4th ed.). Springer.
4. Heiberger, R. M., & Holland, B. (2015). *Statistical analysis and data display: An intermediate course with examples in R, SAS, and SPSS* (2nd ed.). Springer.
5. Hamilton, L. C. (2013). *Statistics with STATA: Updated for version 12* (8th ed.). Cengage Learning.